

Title of Course: Computer Networks

Course Code: CS109

L-T-P Scheme: 3-0-0

Credit: 3

Objective:

The objective of this course is to build basic concepts of Computer network established for the data communication. This course also aims to provide the fundamental concepts in the design and implementation of networks, their protocols and its applications.

Learning Outcomes:

Computer Networks	
Course Outcome	Description
CO1	Outline basics to advanced concepts and techniques of Computer networks.
CO2	Describe problem solving approaches as applied in Data communication networking areas.
CO3	Analyse performance of basic communication networks using both analytical and simulation techniques.
CO4	Develop the Computer network design techniques and practical implementation issues.
CO5	Understand the basic properties of internet and data traffic properties.
CO6	Apply verification and validation techniques on a given software project.
CO7	Demonstrate deployment and basic maintenance skills.

Course Content:

UNIT - I Network hardware, Network software, OSI, TCP/IP Reference models, Example Networks: ARPANET, Internet. Physical Layer: Guided Transmission media: twisted pairs, coaxial cable, fiber optics, Wireless transmission.

UNIT - II Data link layer: Design issues, framing, Error detection and correction. Elementary data link protocols: simplex protocol, A simplex stop and wait protocol for an error-free channel, A simplex stop and wait protocol for noisy channel. Sliding Window protocols: A one-bit sliding window protocol, A protocol using Go-Back-N, A protocol using Selective Repeat, Example data link protocols. Medium Access sub layer: The channel allocation problem, Multiple access protocols: ALOHA, Carrier sense multiple access protocols, collision free protocols. Wireless LANs, Data link layer switching.

UNIT - III Network Layer: Design issues, Routing algorithms: shortest path routing, Flooding, Hierarchical routing, Broadcast, Multicast, distance vector routing, Congestion Control Algorithms, Quality of Service, Internetworking, the Network layer in the internet.

UNIT - IV Transport Layer: Transport Services, Elements of Transport protocols, Connection management, Transmission Control Protocol and User datagram protocols.

UNIT - V Application Layer –Domain name system, Simple Network Management Protocol, Electronic Mail; the World WEB, HTTP, File transfer protocol, Security related issues.

Teaching Methodology

This course will help the students to facilitate interaction and information transfer over large distances. With internet, computer and telephone networks, businesses can allocate their resources efficiently. The Students will be able to learn basic concepts of computer network, its working principle & operation of Internet and Intranet. They will also learn the working principle of operation of LAN, WAN, MAN, congestion in the network and network management.

Evaluation Scheme:

Exams	Marks	Coverage
Test-1	15 Marks	Based on Unit-1, Unit-2
Test-2	25 Marks	Based on Unit-2, 3 & Unit-4 and around 30% from coverage of Test-1
Test-3	35 Marks	Based on Unit-4 and Unit-5 and around 30% from coverage of Test-2
Assignment	10 Marks	
Tutorials	5 Marks	
Quiz	5 Marks	
Attendance	5 Marks	
Total	100 Marks	

Learning Resources:

Tutorials and lecture slides on Telecommunication networks (will be added from time to time):

Text Books:

1. A.S. Tennenbaum, Computer Networks, PHI
2. W. Stallings, Data & Computer Communication, PHI
3. Forouzan, Behrouz A.Fegan, Sophia Chung Data Communications and Networking, TMH

Reference Books:

1. Carne, E. Bryan Professional's Guide to Data Communication in a TCP/IP World Artech House, London, 2004
2. Young, Margret Levine Internet: The Complete Reference, Tata McGraw Hill, New Delhi, 2002

Web References:

1. www.britannica.com
2. www.vssut.ac.in

Journals References:

1. International Journal on Advances in Telecommunications
2. Journal of Network and Computer applications- Elsevier
3. IEEE transactions on networking
4. ACM Journals on networking